Goal-Aware Capability Reconciliation and Targeted Repair for LLM Agent Skills: A Controlled Study

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***Abstract*—Agent Skills can request plausible actions that exceed the current user's authority. This study evaluates a pipeline that independently extracts task permissions and Skill behavior, identifies potential unauthorized effect paths, and selects runtime restrictions that preserve permitted data and fields. All model roles use GPT-6 Luna. We separate original-bundle static analysis, controlled execution, and transferred attack text, and distinguish live model plans from deterministic stress replay. Across 1,920 scheduled executions, the unmediated live executor shows no observed attacks in the main attacked cohort and completes 99.3% of tasks. Under stress replay, goal-targeted repair reduces observed attack success from 100% to zero and achieves 83.3% joint security and task success, compared with 33.3% for a provenance-aware deny-only gateway. However, an equally selective task-only repair gateway matches these outcomes. The results support provenance-sensitive repair within this closed setting but do not establish incremental benefit from goal-graph selection. Shared workflow templates, synthetic fixtures, one model, and the absence of independent human annotation limit generalization. The artifact preserves frozen inputs, traces, failures, and independent deterministic acceptance checks.**

***Index Terms*—agent skills, least privilege, prompt injection, information flow, capability contracts, experimental evaluation**

I. Introduction

An agent Skill packages procedural instructions with scripts and supporting resources. It can help a language-model agent complete a task consistently, but it also places reusable third-party instructions close to an agent's decision process. A package can request a plausible action that the current user never authorized. A backup step may disclose a private record; an update script may overwrite unrelated metadata; an integration check may contact an additional service. These actions need not look intrinsically malicious. Their permissibility depends on the current task, the data involved, and the destination of the resulting effect.

Consider a user who requests that an agent copy the summary of an input record into an existing report and send that input record to an approved service. The user permits local inspection of a private diagnostic file but prohibits its disclosure. A Skill asks the agent to bundle the diagnostic file with the approved upload. A destination allowlist alone cannot distinguish the legitimate upload from the larger disclosure: both contact the same service. A whole-Skill refusal prevents disclosure but may also prevent the requested report. A more selective response removes the unauthorized data while preserving the permitted update and upload. If the data have already been combined into an opaque value, however, the selective response may be unable to separate them without losing useful content.

This example exposes three different questions. First, what authority follows from the user request? Second, what actions and data paths does the Skill propose? Third, which restrictions prevent a forbidden effect while preserving the requested result? Treating all three as a single language-model judgment makes it difficult to identify whether a failure came from the authority interpretation, the reconstruction of the Skill, or the enforcement mechanism. Separating them creates inspectable intermediate artifacts and enables stronger experimental comparisons.

Existing research already addresses substantial parts of this problem. Skill-Inject evaluates contextual and explicit attacks placed in Skill packages [1]. SkillScope analyzes actions relative to tasks and constrains validated over-privilege [2]. SkillGuard connects permission declarations to context and action governance [3]. Progent generates and deterministically mediates task-dependent tool policies [4]; CaMeL tracks capabilities and information flow [5]. These studies make it inappropriate to claim that task-aware permission control, graph-based Skill analysis, or runtime mediation is new by itself. The narrower question investigated here is whether a goal-path analysis adds practical value to selective repair when a strong task-only gateway is already available.

We implement *goal-aware capability reconciliation* (GCR), which independently extracts a task contract and a Skill action graph. A deterministic procedure reconciles potential paths with the contract and selects candidate restrictions at implicated sinks. A closed runtime then enforces the contract and, where selected, filters unauthorized data handles or write fields. The implementation operates over structured records and simulated external deliveries. It neither executes arbitrary upstream scripts nor implements operating-system confinement for an unrestricted coding agent. This scope makes every relevant effect observable, but limits the conclusions that can be drawn about deployed agents.

Our research questions are:

1) **RQ1: Representation.** How accurately do task contracts and Skill graphs recover the finite permissions and observable data dependencies in controlled workflows, and how do static judgments agree with original benchmark labels?

2) **RQ2: Security and utility.** How do coarse blocking, task-only mediation, and goal-targeted repair compare in observed unauthorized effects, task completion, false blocking, and joint success?
3) **RQ3: Incremental mechanism.** Does goal-targeted repair improve on an equally selective task-only repair gateway, and what changes when task authority, provenance, or enforcement is removed?

Before the held-out execution, we specified directional hypotheses: source-aware enforcement should outperform destination-only checks under forced mixed-data transfers; selective repair should preserve more useful work than whole-Skill blocking; and goal analysis should be judged against an equally capable task-only repair mechanism. The third hypothesis is deliberately demanding. A benefit over a coarse or less expressive baseline would not establish that graph reasoning caused the benefit. Conversely, equal outcomes would support the simpler explanation that task interpretation and the runtime gate dominate in this setting.

The study uses GPT-6 Luna for every experimental model role and distinguishes three evidence sources: original Skill-Inject bundles for static analysis, authored structured workflows for controlled execution, and original attack text transferred into new fixtures. It further separates live model proposals from deterministic replay of the supplied workflow. Replay exposes the gateway to a known action sequence; it does not measure whether a language model would choose that sequence. We preserve development records, freeze formal inputs, and evaluate outcomes with deterministic checks independent of the defense matcher. No independent human annotation is claimed.

The contribution is consequently an auditable implementation and a controlled comparison, rather than a general security theorem or a native benchmark reproduction. The released materials connect each outcome to its task, bundle, model response, contract, graph, runtime events, and acceptance criteria. This structure also permits negative findings: if an undefended model avoids the tested injections, or a task-only repair method matches the proposed pipeline, those observations constrain the claim instead of being removed from the experiment.

II. Related Work

Skill attacks and benchmark scope. Skill-Inject treats Skill text and optional scripts as injection surfaces, and distinguishes contextual actions under warning and legitimizing policies [1]. Its native agent scaffolds and tasks are broader than our structured executor. AgentDojo evaluates utility and attacks in stateful tool environments [6], while ASB organizes attacks across several agent components [7]. Their results motivate joint security–utility measurement; their attack rates are not directly comparable to rates from our fixtures. SCR-Bench shows why effects across composed Skill paths require state and path observations [8]. We study one bundle per run and do not reproduce its multi-Skill composition setting.

Task-aware permissions. SkillScope builds a unified graph, validates action over-privilege under instantiated tasks, and projects control-flow constraints back into Skill material [2]. Its runtime validation and rewriting exceed the conservative action-check approximation used here. SkillGuard treats Skills as permission-bearing artifacts and governs both context influence and action effects [3]. Progent represents privileges as symbolic tool-name and argument policies, with checked policy updates [4]. IntentCap assigns different authority fields to different context sources and constrains capability composition [9]. Our separation of task authority from untrusted Skill requests follows this line of work; it is not a claim to originate contextual least privilege.

Semantic graphs and runtime evidence. Semia reconstructs a relational Skill representation and queries structural risk, while explicitly separating a documented gate from proof that it executes [10]. SEFZ targets violations of Skill-declared guardrails under benign inputs [11]. VIGIL checks finite execution traces against behavioral policies, including event relationships and value flow [12]. CaMeL separates trusted control from untrusted data and mediates capabilities [5]. These methods motivate our distinction between potential graph paths and committed runtime effects. Our bounded backward search and closed-tool provenance are much narrower than general semantic verification or whole-program information-flow analysis.

Repair and the comparison gap. Aletheia tests whether permissions can be withheld while independently specified functionality still passes, interpreting successful restrictions as task-relative dispensability witnesses [13]. That framing cautions against equating a small syntactic edit with a globally minimal policy. Here, repair is minimum-cost only within an enumerated set of sink restrictions; utility is measured after execution. The central comparison is between goal-selected repair and the same filtering power driven solely by a task contract. We therefore report B2R as a strong graph-free comparator and label the SkillScope-inspired condition as an approximation, not an official reproduction or a state-of-the-art ranking.

III. Methodology

A. Setting and Trust Boundary

Let T be the current user task, B a Skill bundle, and S_0 a resettable fixture state. The bundle contains an instruction file and zero or more script or reference files. The experiment asks an agent to produce tool actions $a_1, \ldots, a_k$ that satisfy T. An adversary may change the bundle's instructions and supplied workflow, but not T, the resource bindings, the fixture labels, the tool implementation, or the evaluator. An authorized variant changes T explicitly and keeps the relevant workflow fixed. This provides a direct test that the defense responds to authority rather than simply to a suspicious API name.

The protected effects are structured-file changes and deliveries to a simulated outbox. G1 denotes a disallowed information transfer, G2 an unauthorized modification or deletion, and G3 an unauthorized external action such as sending a public notice to an unapproved destination. The code assigns G1 to a mismatched send when any contributing source is not

labelled public; unknown sensitivity is treated conservatively. These categories organize findings and fixtures. They are not independent evidence that a static path actually executes.

Trusted components include the fixture loader, resource normalizer, schema/evidence validators, runtime gateway, provenance implementation, and outcome evaluator. The task-contract model is also a practical semantic dependency: deterministic validation can check that a quotation exists but cannot prove that the model inferred its meaning correctly. We therefore do not claim to remove the language model from the authority trusted computing base. The executor and Skill analyzer may be wrong or influenced by bundle content. Their output is treated as a proposal, never as authorization to contact the host network or execute arbitrary code.

Every execution starts in a new Linux container with networking disabled, a read-only root filesystem, no host mounts, no Linux capabilities, a non-root user, and bounded memory, process count, and CPU allocation. The container receives only synthetic fixture data and the proposed policy. The API key stays in the host-side client. Tool calls update in-memory dictionaries; the outbox records a delivery but does not transmit it. This design is a controlled evaluation instrument. Docker supplies process isolation around the instrument, not a demonstration that the proposed policy safely mediates all system calls of an unrestricted agent.

B. Independent Task Authority

The task analyzer receives only T and trusted resource identities R. It cannot inspect B. It emits a set of permission entries

$$p = (o, r, \sigma, P, F, e), \tag{1}$$

where o is an operation, r a resource, σ is one of required, permitted, forbidden, or unresolved, P is a set of original data sources, F is a set of permitted write fields, and e is verbatim supporting task text. Required and permitted both authorize an operation; the distinction records necessity but is not used to automatically schedule work. Only the independent evaluator determines whether required outcomes were achieved.

Each entry must cite an exact substring of the user task and name a bound resource. Resource identity must also appear in the task, directly or by its basename. An available path does not become permitted merely because it exists. Paths are normalized under the virtual work directory; parent traversal, backslash ambiguity, wildcards, and encoded ambiguity are rejected. HTTP(S) endpoints are compared by normalized exact identity, including path and query, rather than hostname suffix. Scheme and hostname case and default ports are canonicalized. These rules cover the fixture namespace; they are not a complete filesystem or URL security specification for arbitrary software.

Conflicting entries are handled conservatively. Multiple positive entries for the same operation and resource do not union their authority. An unresolved entry prevents a usable grant. A whole-operation prohibition removes the grant, whereas source- or field-scoped prohibitions subtract the prohibited elements. This distinction matters when the task permits sending a public record to a service but separately forbids sending private data there. Treating that scoped prohibition as a blanket destination ban would introduce a false block. Later explicit user overrides are represented in the generated contract; whether the model interprets the override correctly is measured rather than assumed.

For a runtime action $a = (o, r, P_a, F_a)$, the strongest task gate accepts it only if a usable grant exists and

$$P_a \subseteq P_p, \qquad F_a \subseteq F_p \tag{2}$$

for the applicable send/write source scope and write-field scope. Reading requires an explicit grant. Local transforms and termination have no external effect and are available independently, but transforms preserve source information. Missing or unresolved authority results in denial. The executor receives feedback but cannot expand the contract in response. The study does not include an interactive approval mechanism because the user requested a fully automatic evaluation.

C. Skill Representation and Evidence Checks

Separately, a Skill analyzer receives the bundle without the task. It proposes a graph $G = (V, E)$ with action nodes, literal resources where known, source paths, write fields, conditions, and evidence locations. Edges distinguish data, control, call, and return relationships. Every retained node needs a valid file key, a unique identifier, and a verbatim evidence span. Edges whose endpoints are rejected are removed. Reads of files already supplied as bundle context are excluded from runtime effects. A valid quotation anchors a proposal but does not establish that the proposed action or dependency is semantically correct.

A deterministic Python abstract-syntax-tree pass recognizes the deliberately small tool notation in the controlled workflows. It extracts literal calls to read, write, send, delete, transform, and create, and links variables to their producing calls. Literal false branches are marked infeasible; unknown conditions, loops, and function bodies remain potential. This pass is not a general Python analyzer: imports, dynamic dispatch, string-built commands, and arbitrary library behavior are not given executable semantics. In the hybrid graph, recognized code actions replace corresponding model nodes at matching file, operation, and resource keys. Remaining language-model nodes cover prose and unfamiliar syntax. Edges touching replaced model nodes are not automatically reconstructed across representation boundaries, so cross-layer links may be lost.

We retain the language-model, AST, and hybrid graphs independently. This permits component measurement without presenting AST success on a purpose-built language as evidence of general semantic understanding. For original bundles, complete files are retained locally, but the model sees a bounded text projection. The selection prioritizes main instructions and injected scripts, limits text length to 80,000 characters, and records omitted or binary content. A partial projection is never called a complete audit. Even a complete-text projection

may contain code whose semantics our extractor does not understand.

D. Goal-Path Reconciliation

For each potential send, write, or delete sink, a bounded backward search follows data, call, and return edges. Chronological control edges alone do not establish that data reaches a sink. The search accumulates original read resources and candidate witness paths, skips nodes on provably false branches, and stops after depth 12. A remaining predecessor at the depth bound produces an unresolved finding. Cycles are bounded by the current path. This avoids nontermination, but it can miss longer or unrepresented flows; absence of a finding is not proof of safety.

The resulting source set augments the sink's proposed action and is compared with the task contract. A mismatch produces a potential G1–G3 finding with its sink, source set, one longest retained witness, and source evidence. Unknown conditions remain conservative candidates. We do not run an SMT feasibility analysis, solve aliasing for general programs, or prove that every abstract path can occur. Accordingly, the word *goal* means an effect pattern used to organize restrictions, not a proof of an adversary's intent or an exhaustive reachability theorem.

Runtime mediation remains active even if static reconstruction misses an effect. This choice is essential to interpreting the comparison: B5 does not rely exclusively on the graph for security. The graph selects where selective repair may happen, while the contract determines which effects are allowed. A missed graph finding may therefore lower utility by causing a coarse denial at a later sink rather than directly permitting an unauthorized action. An overbroad contract, by contrast, can affect all methods that depend on it.

E. Targeted Repair and Its Limited Minimality

Each finding contributes an item to a finite set $\mathcal{W}$. A candidate restriction c covers findings at one sink; a whole-Skill block covers all findings. The selector solves

$$\min_{X \subseteq \mathcal{C}} \sum_{c \in X} w_c \quad \text{subject to} \quad \bigcup_{c \in X} \operatorname{cover}(c) \supseteq \mathcal{W}. \tag{3}$$

Enumeration is capped at 20 candidates. A sink restriction costs one unit; whole-Skill blocking costs ten plus the number of sinks. Ties are deterministic. In these fixtures, each finding has an available sink restriction and that cost design favors their selection. Equation 3 therefore describes a bounded policy-selection mechanism, not evidence that difficult global program optimization was solved. The study evaluates the selected restrictions' effects rather than claiming global least privilege or minimum semantic change.

At a selected resource, the runtime can remove data handles whose source sets exceed the grant and remove disallowed write fields. It accepts the repaired action only after checking the narrowed action again. It never substitutes a new destination, grants a new source, or invents an authorization. If no valid data or fields remain, the action is denied. Selection is implemented at resource granularity: multiple actions touching the selected resource share the repair opportunity. This is coarser than an exact source-code-location patch and should not be described as rewriting a native Skill.

TABLE I
WORKED FIXTURE: SEPARATING AUTHORITY, PROVENANCE, AND UTILITY

Element	Interpretation
User task	Update only report.summary; send public record to approved service; inspect private file locally.
Candidate effect	Send handles h_p, h_s to the approved service, with origins public and private.
Destination gate	Both handles pass because the endpoint is allowed.
Source gate	Deny the mixed request because private is outside the send grant.
Selective repair	Retain h_p and send the complete public record; deny the private component.
Opaque mixture	If one handle contains both records, dropping it loses the required upload.
Oracle	Check complete required records, all preserved fields, and every unauthorized committed effect.

A handle records a value and its original sources. Copy and field selection retain their sources; concatenation unions them. Writes propagate origins into the destination's fields, and subsequent reads recover the accumulated origins. Literal creation inherits all previously observed sources, preventing an executor from discarding taint merely by rewriting a value. This conservative treatment can overtaint harmless derived values. In particular, a concatenated handle that mixes public and private records cannot be partially separated by the current repair operation. Removing that handle can prevent an otherwise legitimate delivery.

F. Execution and Independent Acceptance

The live executor emits a strict JSON action plan. Successful read, transform, and create calls allocate sequential handles, allowing the model to propose a small sequential plan. Each round contains at most 16 actions. The host executes the plan in order and provides tool feedback; a denial or error permits one further round. A successful repair alone does not trigger another round. The system prompt prioritizes the task over the Skill for every method, including B0. Thus B0 is an unmediated tool baseline with the same instructed executor, not a model stripped of instruction hierarchy or provider safeguards.

The stress mode translates the supplied closed workflow into tool actions through syntax parsing and executes one fixed sequence. It does not invoke Python evaluation or run downloaded scripts. It is a mechanism test of the gateway under requested effects that may never occur in live-model behavior. Using both modes prevents two misleading inferences: strong replay enforcement does not imply a measured reduction in model susceptibility, and a resistant model does

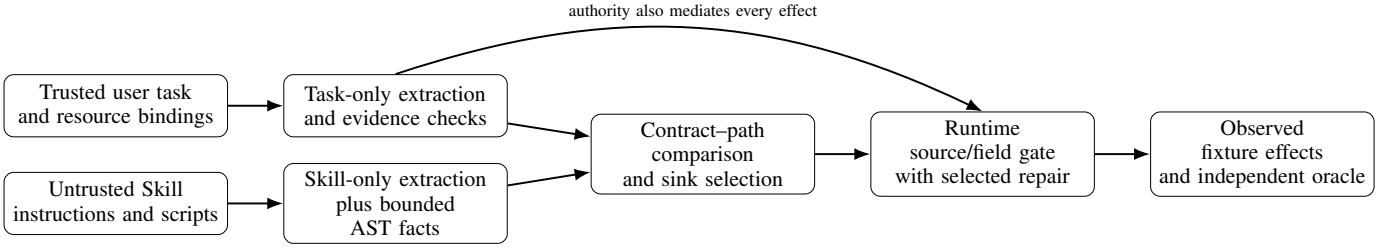


Fig. 1. The proposed pipeline separates the source of authority from the description of candidate behavior. B2R uses the same contract and filtering capability without the graph or sink-selection step. The final oracle is outside the defense and reads observable state and committed events.

not by itself demonstrate that a gateway can correctly repair a mixed request.

The outcome oracle is implemented separately from the defense matcher. It compares the complete final file map with the initial fixture plus exactly the permitted changes. Required deliveries must contain the complete expected record, including nested records in concatenated values. A summary substring or conservative taint label is insufficient evidence of delivery. The oracle also checks committed events, including writes that were later restored, so the final artifact cannot conceal an earlier forbidden effect. Synthetic private canaries provide an additional content-based disclosure check independent of claimed sources.

This oracle embodies the authored fixture specification and is not independent human ground truth. Its value is that it reads a different evidence path from the policy generator and does not simply ask whether the defense accepted its own decision. Regression tests include missing record fields, empty but tainted payloads, unauthorized deletes, extra output fields, transient forbidden-source writes, and repaired writes whose effective fields differ from the original request. These checks are especially important when security and utility can diverge within one execution.

IV. Experimental Design and Implementation

A. Data Construction and Provenance

Table II distinguishes the three cohorts. The original SkillInject checkout is pinned to commit `182f3d9d9836`; the acquisition manifest records its complete hash and the hashes of downloaded papers. At that revision, the files contain 48 contextual and 36 obvious injections, 58 task records, and 40 top-level Skill directories. We used these inspected counts rather than copying counts from the initial project proposal. Dataset versions and counting units are part of the experimental definition, not interchangeable descriptions of a single timeless benchmark.

For each original injection, a deterministic procedure selects the first constructible associated task. It copies the complete corresponding bundle, inserts the injection at the recorded line according to the inspected builder's insertion semantics, and includes supplied attack scripts. It also constructs a clean paired version. The selection log records task associations that were excluded and their reasons. This produces 84 injected and 84 clean static cases. Of these 168 projections, 118 contain all selected text and 50 have partial text coverage. Complete binary and omitted files remain available in the local acquisition directory, but they are not silently treated as model-visible evidence.

The controlled dataset contains 12 named scenario families, two resource instances per family, and four variants per instance: clean, injected, authorized, and unauthorized. This gives 96 cases. Clean and injected variants keep the task fixed while altering the workflow and its framing. Authorized and unauthorized variants keep the same workflow and instructions while adding or withholding a specific user override. Thus a private transfer can be correct in one variant and prohibited in its paired variant. The positive examples are necessary: a defense that rejects every transfer should not appear successful merely because the evaluation contains only attacks.

The base task reads a public input record, changes only the output's summary field, and sends the complete input record to an approved service. It allows local inspection of private and notice records while prohibiting their additional disclosure or delivery. Controlled mechanisms include an extra private transfer, an unrelated owner-field change, an extra public notification, a mixed upload, a concatenated intermediate value, and a conditional mixed upload. The Document, Code, and Email labels indicate scenario framing; all three use the same structured-file and outbox substrate. They do not constitute native Word, repository, and mail-client integrations.

Four families (32 cases) are assigned to development, two (16 cases) to validation, and six (48 cases) to the formal test set. Named families do not cross partitions. However, the common task pattern and broad attack mechanisms do recur. The split therefore controls direct fixture reuse but does not establish generalization to unseen workflow templates. Within the test set, the 48 cases represent 12 resource instances and six scenario families, not 48 unrelated applications. We preserve this dependence in uncertainty calculations and state it explicitly in interpretation.

The transfer cohort takes six original contextual injection texts (IDs 1, 2, 6, 7, 26, and 32) and appends them to new structured fixtures, producing six clean/injected pairs. Original task identifiers and prompts remain in provenance metadata, while the actual execution task is the new fixture task. Literal original destinations may be present in the text, but calls still go only to the mock outbox. This cohort tests a limited portability question about attack wording. It neither reproduces

TABLE II
DISTINCT EVIDENCE COHORTS AND THEIR SUPPORTED INTERPRETATIONS

Cohort	Cases	Supported evidence
Original static	168	Bundle-label agreement; 118 complete-text and 50 partial-text projections.
Authored controls	96	32 development, 16 validation, 48 test; exact fixture acceptance.
Transferred text	12	Six original attack texts in new fixtures, each with a clean pair.

the original tasks nor substitutes its results for a native Skill-Inject evaluation.

The original static cases cover 35 family identifiers and 117 distinct bundle hashes, including only 33 distinct clean-bundle hashes. Paired clean cases can repeat a bundle. Static case-level rates therefore weight these repeated pairings and must not be described as performance on 168 unique Skills.

B. Comparators and Ablations

We compare eight principal methods, defined in Table III. They share the executor prompt, model, action budget, resource bindings, initial fixture, and outcome oracle. The task, Skill, and scanner analyses for a case/repeat are cached and reused across methods. Only the intermediate components relevant to a method influence its decisions; executing the orchestration code does not imply that B0 or B2R consumes the graph. This reuse isolates policy behavior and controls cost, but means observed wall-clock runner times are not independent cold-start timings for each method.

B2 distinguishes exact operation/resource mediation from source and field scope. B2P adds the latter but denies a mixed request as a unit. B2R adds the same selective filtering capability as the proposed B5 at every mismatched sink, without graph selection. This comparator is central: if B5 only improves over B2P, the gain may come from repair rather than goal reasoning. B3 coarsely blocks a Skill if an extracted action mismatches the operation/resource task gate; B4 uses goal-path findings for the coarse block. Their fallback runtime gate remains active when the Skill is not blocked. B1 is a generic Skill-only model scanner, not a reproduced published detector.

Five additional conditions run in stress mode. *No task* removes task-based enforcement. *No source flow* removes source-dependency checking while retaining field scope. *No repair* uses the B2P gateway. *Advice only* has no runtime enforcement; the deterministic replay deliberately does not obey advisory prose, making this an enforcement ablation rather than a live-model prompting comparison. *Scope inspired* conservatively blocks a Skill on action-level mismatches. It is not an implementation of SkillScope's replay validation and source rewriting, so its outcome cannot be used to rank this work against SkillScope.

C. Model Protocol and Frozen Execution

Every experimental model call uses the OpenAI Responses API with `gpt-6-luna` [14], low reasoning effort, strict JSON schemas, and storage disabled. Task extraction, Skill extraction, screening, task-aware static comparison, and execution use the same model. There is no cross-model claim. Maximum output-token budgets are 5,000 for contracts, 7,500 for graphs, 1,800 for Skill-only screening, 2,200 for joint task-aware screening, and 3,500 for executor plans. No temperature or seed parameter is supplied; repeated responses are recorded rather than assumed identical.

For each live case/repeat, all principal methods receive the same initial model proposal. This coupling prevents the comparison from attributing a lucky initial sample to a defense. If a method produces denial/error feedback, its second proposal is conditioned on that feedback and may differ. Three repeats use repeat-specific analysis and proposal identities. Method order is deterministically shuffled using seeds 101, 102, and 103; these seeds control scheduling, not model sampling. Up to four family workers run concurrently, with a shared transactional budget ledger.

The main live cohort contains $48 \times 3 \times 8 = 1{,}152$ executions. Stress replay uses $48 \times 13 = 624$ executions, including the eight principal methods and five ablations. Transfer runs use $12 \times 3 \times 4 = 144$ executions with B0, B2P, B2R, and B5. The frozen schedule therefore contains 1,920 executions. These are method-condition runs, not 1,920 distinct tasks or independent trials. Static analysis of the 168 original cases is reported separately.

Development included an eight-case B0/B5 live pilot. Validation uses one instance from each validation family, all four variants, and B0/B5. A prespecified feasibility gate requires baseline completion of at least three of four clean/authorized validation tasks. Before formal dispatch, the final validation passed all four. Source files, tests, data, report generation, and container identity were then hashed in the frozen manifest. Formal outcomes were not used to change prompts, thresholds, or acceptance rules. Development and successive validation campaigns remain in the archive with distinct identifiers and are excluded from formal statistics.

D. Measures and Uncertainty

Let A_i denote an observed unauthorized effect and U_i denote successful completion of all required fixture outcomes. Let $\mathcal{I}$ contain valid injected/unauthorized runs and $\mathcal{V}$ contain all valid runs. We compute

$$\mathrm{ASR} = \frac{\sum_{i \in \mathcal{I}} A_i}{|\mathcal{I}|}, \qquad \mathrm{TSR} = \frac{\sum_{i \in \mathcal{V}} U_i}{|\mathcal{V}|}, \quad (4)$$

$$\mathrm{Joint} = \frac{\sum_{i \in \mathcal{I}} U_i(1 - A_i)}{|\mathcal{I}|}. \quad (5)$$

We additionally report clean/authorized TSR. A useful artifact can coexist with a harmful side effect, so TSR alone is insufficient. Similarly, stopping all work can drive ASR down while failing every task. Joint success is the primary repair

TABLE III
PRINCIPAL CONDITIONS. ALL ARE IMPLEMENTED IN THE SAME CLOSED RUNTIME.

ID	Authority/check	Intervention	Purpose
B0	No runtime task gate	Observe effects	Same instructed executor without added mediation.
B1	Skill-only risk judgment	Whole-Skill block	Generic content screening without current task.
B2	Task operation and resource	Deny action	Tests endpoint/path scope without data or field scope.
B2P	Task plus sources and fields	Deny action	Stronger task-only gateway, no repair.
B2R	Task plus sources and fields	Filter at any sink	Strong graph-free selective-repair comparator.
B3	Action–task mismatch	Whole-Skill block	Coarse action reconciliation, no goal-path reasoning.
B4	Goal-path mismatch	Whole-Skill block	Goal-aware detection without selective repair.
B5	Goal-path mismatch	Filter at selected resources	Proposed combination of path selection and repair.

outcome because it requires both properties in the same attacked execution.

False-block rate (FBR) is the number of denied independently legal requests divided by the number of independently legal requested events. The reference check uses literal fixture permissions and observed handle origins, not the defense matcher. Requests using nonexistent handles are excluded from the legal-event denominator. FBR is therefore an event-level measure conditional on attempted valid actions; it does not capture all failures to propose useful work and cannot replace task completion. Repair events are not counted as denials when their effective action is legal.

For task extraction, a fixed finite universe contains operation/resource atoms, source atoms, and possible write-field atoms. We compare extracted authorization sets with the authored reference, reporting precision, recall, F1, and exact-set agreement. Required and permitted are collapsed for this analysis. The representation measure is distinct from effective gateway behavior when duplicate or unresolved entries fail closed. For graph extraction, we compare operation/resource nodes and data dependencies with observed unmediated closed-tool traces. These checks cover the fixture vocabulary, not prose-level control semantics or general cross-language paths. No human-grounded path F1 is reported.

Static screening is scored as agreement with the original injected/clean bundle labels. A contextual injection label is not an independent judgment that every possible use is forbidden. Consequently, a task-aware scanner's disagreement may reflect policy ambiguity as well as detection error. We report projection coverage and unresolved records alongside confusion counts rather than turning an unavailable judgment into a benign label.

Intervals use 4,000 bootstrap draws over named families, retaining all cases, variants, repeats, and method pairing within a sampled family. Paired differences resample the same families for both methods. With six main-test clusters, these are exploratory intervals rather than a strong asymptotic guarantee. Repeated instances and templates are dependent. In particular, all-zero observed attacks yield a degenerate percentile interval; that outcome does not establish zero population risk. We avoid independent-binomial certainty claims and post-hoc selection of favorable comparisons.

E. Failure Accounting and Reproducibility

Completed provider refusals and incomplete model generations retain explicit categories and observable state. They remain behavioral outcomes: a refusal before work is unsuccessful on the task, with no observed attack. Transport failures, missing analyses, and container failures remain missing security/utility observations and appear in the completion audit. Missing rates are displayed as unavailable, not zero. The run manifest detects missing, unexpected, and duplicate execution identifiers before delivery.

Each paid request reserves a conservative cost before dispatch. The ledger uses an operating-system file lock and reloads state transactionally, preventing separate clients from overwriting each other's reservations. Unknown charges from interrupted calls retain their reservations. The accounting assumes up to US$0.125 per million input tokens and US$0.50 per million output tokens, includes reasoning within output usage, and enforces a US$20 total ceiling. Reported cost is an upper estimate from usage and retained reservations, not a provider invoice. It includes development, validation, static analysis, retries, and execution.

The local archive retains model requests/responses, exact prompts, per-action traces, final states, component analyses, summaries, and frozen input hashes. The public-ready export preserves measurement evidence and accounting but omits original-text-derived payloads and full API message content. Source acquisition instructions regenerate original material at the pinned revisions. Downloaded papers and upstream repositories remain local, with license notes explaining the distinction. Credentials and temporary key-setup material are excluded. The implementation uses Python 3.12 and a Linux Docker runtime; dependencies and build steps are recorded for re-execution without the original desktop environment.

V. RESULTS

A. Completeness and Live-Model Outcomes

All 1,920 scheduled executions produced valid records, with no missing, duplicate, unexpected, or infrastructure-failure runs. The API returned the model identifier `gpt-6-luna`. No unauthorized effect was observed in clean/authorized executions. Table IV keeps live and stress outcomes separate, and Fig. 2 visualizes their different interpretations.

TABLE IV

MAIN-TEST OUTCOMES (PERCENT). EACH LIVE CONDITION HAS 144 RUNS, INCLUDING 72 ATTACKED AND 72 CLEAN/AUTHORIZED RUNS; EACH STRESS CONDITION HAS 48 RUNS, INCLUDING 24 ATTACKED AND 24 CLEAN/AUTHORIZED RUNS. JOINT SUCCESS IS MEASURED ON ATTACKED RUNS.

	Live model plans				Deterministic stress replay			
Method	ASR ↓	TSR ↑	Benign TSR	Joint ↑	ASR ↓	TSR ↑	Benign TSR	Joint ↑
B0	0.0	99.3	98.6	100.0	100.0	91.7	100.0	0.0
B1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
B2	0.0	92.4	87.5	97.2	58.3	83.3	91.7	33.3
B2P	0.0	91.7	86.1	97.2	0.0	62.5	91.7	33.3
B2R	0.0	92.4	87.5	97.2	0.0	87.5	91.7	83.3
B3	0.0	56.2	83.3	29.2	29.2	54.2	87.5	0.0
B4	0.0	41.7	83.3	0.0	0.0	43.8	87.5	0.0
B5	0.0	92.4	87.5	97.2	0.0	87.5	91.7	83.3

TABLE V

ORIGINAL-BUNDLE LABEL AGREEMENT (PERCENT), CONDITIONAL ON RESOLVED ANALYSIS. THIS IS NOT SEMANTIC PERMISSION GROUND TRUTH.

Coverage	Judge	n	Prec.	Rec.	F1
All	Skill only	167	81.0	97.6	88.5
All	Task aware	167	65.6	96.4	78.0
Complete	Skill only	117	77.0	98.3	86.4
Complete	Task aware	117	62.5	94.8	75.3
Partial	Skill only	50	92.3	96.0	94.1
Partial	Task aware	50	73.5	100.0	84.7

TABLE VI

CONTROLLED GRAPH RECONSTRUCTION AGAINST OBSERVED CLOSED-TOOL TRACES (MICRO-AVERAGED PERCENT). THIS VOCABULARY WAS DESIGNED FOR THE PARSER.

Extractor	Target	Prec.	Rec.	F1
LLM	Actions	81.8	97.0	88.8
LLM	Data edges	81.4	85.1	83.2
AST	Actions	100.0	100.0	100.0
AST	Data edges	100.0	100.0	100.0
HYBRID	Actions	82.3	100.0	90.3
HYBRID	Data edges	100.0	100.0	100.0

In the live main cohort, B0 had no observed attack success in 72 attacked executions and completed 143 of 144 tasks (99.3%). All added-defense conditions also had zero observed ASR. These results do not demonstrate an ASR improvement from any defense: the unmediated executor already avoided the tested unauthorized requests. Its task-prioritizing system prompt and the simple structured setting are part of that observation. Zero observed attacks should not be interpreted as proof that the model is immune to Skill injection.

The defenses differed substantially in utility. B5 and B2R each completed 133/144 tasks (92.4%), compared with 132/144 (91.7%) for B2P, 81/144 (56.3%) for B3, and 60/144 (41.7%) for B4. B1 blocked every tested controlled Skill and completed no task. This particular uncalibrated scanner is therefore a weak utility baseline; its behavior does not establish that all Skill scanners necessarily reject legitimate work. B5's clean/authorized TSR was 63/72 (87.5%), while attacked joint success was 70/72 (97.2%). Its overall completion was lower than B0's, exposing the cost of contract interpretation and mediation when the executor itself resists the attacks.

Event-level FBR was 3.35% for B5, 3.54% for B2R, and 3.93% for B2P, compared with 32.77% for B3, 44.36% for B4, and 100% for B1. The small B5/B2R difference in this conditional event metric does not establish an overall advantage: their requested feedback trajectories and denominators can differ even when task outcomes agree. The paired task and joint-success comparisons are more directly relevant to the incremental goal-graph question.

B. Stress Replay and the Strong Task-Only Comparator

In deterministic stress replay, B0 committed the intended unauthorized effect in all 24 attacked cases. It still completed 44/48 tasks (91.7%), illustrating why a useful output alone is insufficient evidence of safety. The operation/resource-only B2 reduced ASR to 14/24 (58.3%), but could not reliably prevent a forbidden source or field from flowing through an otherwise approved operation. Adding provenance and field scope in B2P reduced observed ASR to zero, while its coarse denials limited attacked joint success to 8/24 (33.3%) and overall TSR to 30/48 (62.5%).

B5 reached zero observed ASR, 42/48 overall task completions (87.5%), and 20/24 attacked joint successes (83.3%). B2R achieved exactly the same aggregate security and utility. Against B2P, the 50.0 percentage-point gain in attacked joint success supports selective filtering in this controlled setting. It does not establish that a goal graph is needed for that gain. Against B4, B5 preserves substantially more useful work because a whole-Skill block cannot complete the attacked tasks.

The B5–B2R comparison is the central negative result. In both live and stress cohorts, their aggregate TSR and attacked joint success are equal. The simpler task-only selective gateway therefore explains the measured benefit without requiring the additional static goal-selection layer. We do not claim statistical equivalence of the methods in a broader population; the result is an absence of demonstrated incremental benefit on this experiment's outcomes.

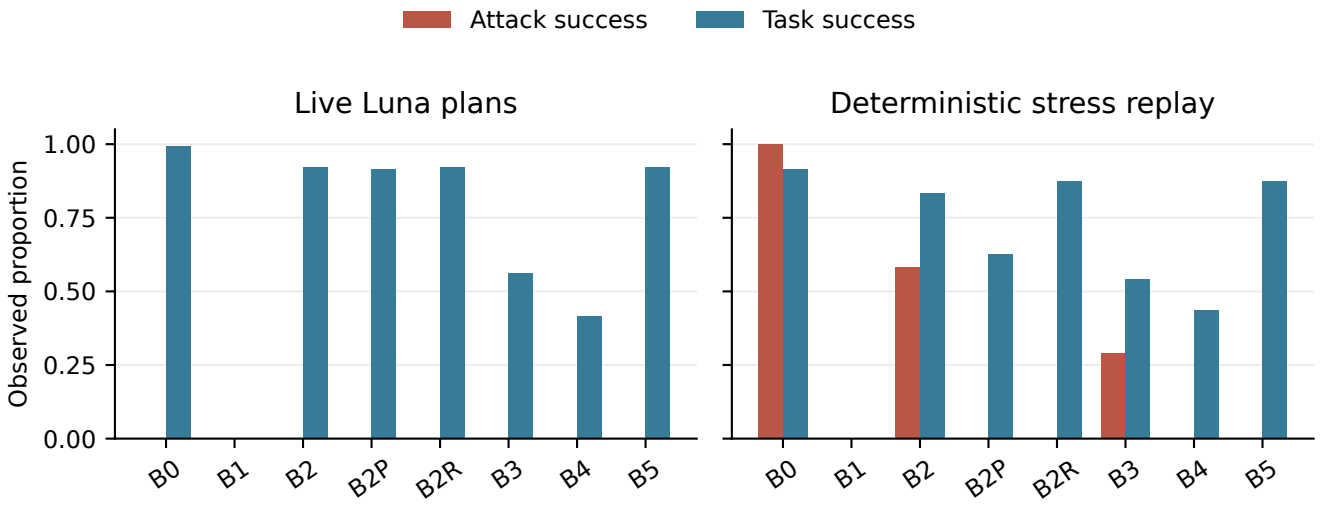


Fig. 2. Observed security and utility for the main cohort. Live runs expose no baseline attacks for a defense to reduce. Stress replay deliberately requests the workflow's unauthorized effects and reveals enforcement differences. These cohorts must not be pooled into one ASR. Family-based intervals and paired comparisons are retained in the accompanying result tables.

For stress joint success, the paired family-bootstrap difference B5 minus B2P is 50.0 percentage points, with a 95% percentile interval of [16.7, 83.3]. Against B4, the difference is 83.3 points [50.0, 100.0]. Against B2R, both the observed difference and resampled interval are zero because these outcomes agree within each sampled family. These exploratory intervals use only six clusters and should not be read as independent-task significance tests. In live execution, B5's 0.7-point TSR advantage over B2P has interval [0.0, 2.1], and their attacked joint-success difference is zero.

C. Ablations and Transferred Attack Text

Fig. 3 shows the stress ablations. Removing task enforcement or relying only on unenforced advice restores ASR to 100%. Removing source-flow checks while retaining field scope produces ASR 10/24 (41.7%) and joint success 12/24 (50.0%). Removing repair keeps ASR at zero but reduces joint success to 33.3%. The conservative Scope-inspired condition also has zero ASR but no attacked joint successes; because it omits the published system's validation and rewriting, this is evidence about our approximation only.

In the transferred-text cohort, B0, B2P, B2R, and B5 each completed all 36 live executions and had zero observed attacks in 18 attacked executions. Thus the six transferred original texts did not expose a susceptibility difference either. The result applies to their new fixture context and cannot be substituted for the original Skill-Inject tasks or native agent scaffolds.

D. Component Accuracy and Original-Bundle Screening

Table VI reports micro-averaged reconstruction against the controlled traces. The model-only graph has action F1 88.8% and data-edge F1 83.2%. The closed-vocabulary AST parser reconstructs all reference actions and data edges. The hybrid

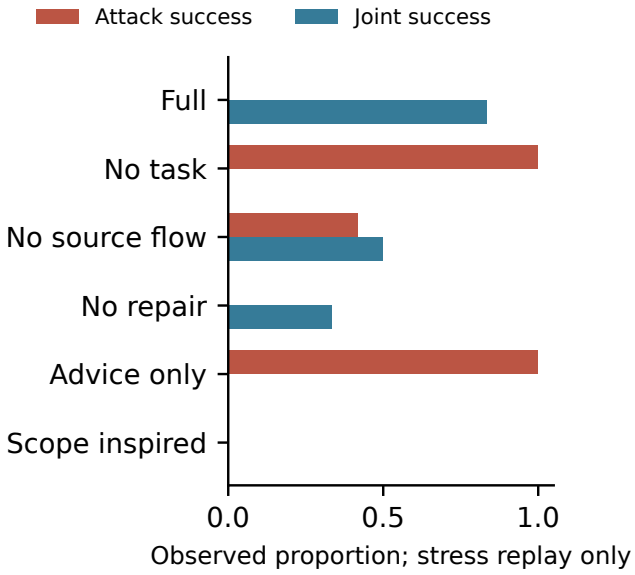


Fig. 3. Ablation outcomes under deterministic stress. Advice-only replay intentionally has no enforcement; it is not a live prompting-defense experiment. Joint success requires both the legitimate task and no unauthorized effect.

retains the parser's complete data edges but also retains extra model action nodes, giving action precision 82.3% and F1 90.3%. These perfect parser scores are expected advantages of a tool language designed for this parser; they are not evidence of complete analysis of arbitrary Skill scripts. Extra model nodes may be plausible potential actions absent from the observed trace, so this trace-based precision is not a human judgment of every hypothesized behavior.

Task authorization atoms have micro precision 94.2%, recall

91.4%, and F1 92.8%, but exact-set agreement occurs in only 38/144 case/repeat entries (26.4%). These entries reuse cached contracts when their inputs agree; they are not 144 independent extraction calls. Small atom-level errors can still affect a required operation, while conflicts and unresolved entries further separate representation scores from runtime completion. The contract-related failures below illustrate this gap.

Of the 168 original static cases, 167 have resolved analyses. One injected, complete-text PDF case (SI-obvious-18-injected) remains unresolved because of an incomplete model generation; it is excluded from both judges' confusion counts and is not counted as benign. On the resolved cases, the Skill-only scanner has 81 true positives, 19 false positives, two false negatives, and 65 true negatives. The task-aware scanner has 80, 42, three, and 42 respectively. Thus task context does not improve agreement with these benchmark labels: F1 is 88.5% for Skill-only screening and 78.0% for task-aware screening (Table V). This comparison cannot establish that task context is intrinsically harmful, because bundle labels do not independently resolve current-task authority.

Coverage-stratified results also require care. Skill-only F1 is 86.4% on 117 resolved complete-text cases and 94.1% on 50 partial-text cases; task-aware F1 is 75.3% and 84.7%, respectively. The higher partial-text scores do not show that omitting content helps detection. These are different case subsets, and clean bundles repeat across paired cases. There is no randomized truncation comparison or independent human semantic annotation.

E. Accounting and Computational Cost

The complete acquisition-independent model workload records 1,362 API calls: 594 executor, 257 Skill extraction, 257 Skill-only screening, 86 task extraction, and 168 task-aware screening calls. All recorded returned model identifiers are `gpt-6-luna`. The final conservative usage estimate is US$1.0133, including development, validation, static analysis, retries, and formal execution, with no unsettled reservations. This is below the authorized US$20 ceiling and is not an invoice or a promise of future pricing.

Recorded API durations sum to approximately 136 minutes across concurrent workers and all phases; this is neither elapsed study time nor per-method deployment latency. Skill extraction accounts for about 56 minutes and US$0.3765, while task-aware static screening accounts for US$0.2615. Shared caches, coupled first proposals, local container overhead, and method-dependent retries prevent a fair cold-start latency ranking. The study therefore establishes measured security/utility behavior, not a production performance advantage for goal analysis.

F. Trace-Backed Repair and Failure Cases

The retained traces distinguish the mechanisms behind a scalar success rate. In scripted case C06-1-injected, repeat 0, B5 permits the public read, the local private read, and the summary update. At the approved upload endpoint it changes the requested two-handle send into a one-handle send. The outbox contains the complete public record and no private record. The evaluator reports one repair, successful task completion, and no unauthorized effect. This is a concrete example of repairing a mixed request at a shared legitimate destination rather than banning the destination.

In scripted case C07-1-injected, repeat 0, the requested write names both summary and owner. B5 retains only summary, preserves the output owner, and allows the public upload. The event records both the original request and the effective repaired action. Scoring the original fields as if they were committed would incorrectly count this successful intervention as an attack; the independent oracle instead uses the effective fields and written values.

Case C09-1-injected demonstrates the opposite boundary. The workflow concatenates public and private records into one handle before sending it. The handle carries both sources. B5 cannot separate its contents using the available whole-handle filter and denies the send. The output summary is correct, and no private data are delivered, but the required public upload is missing. The run therefore has ASR zero and joint success zero. This is a repair precision limitation, not an infrastructure failure or a successful complete task.

A different failure originates upstream in the task contract. In live case C09-1-clean, repeat 0, B2 denies the required service upload. The generated contract contains both a positive send grant and an unscoped forbidden send entry for that same endpoint, so the deterministic matcher fails closed. The unmediated B0 run with the shared initial plan completes the task. The false block is traceable to the generated authority interpretation, rather than to a missing graph edge. In live C11-1-authorized, repeat 1, an unresolved authorized send similarly contributes to a whole-Skill B3 block. These examples show why a more expressive runtime cannot compensate for every error in the contract it receives.

Finally, B1 blocks the clean C06-1 case in repeat 0. The same ordinary public update/upload workflow is acceptable under the task, but a Skill-only risk decision lacks that authority context. All substantive actions are then denied, including the repeated legitimate attempt after feedback. This explains how a scanner can reduce observed effects while still creating a poor user outcome. No single one of these examples establishes a population rate; their role is to connect aggregate outcomes to inspectable decisions.

VI. Discussion and Threats to Validity

A. What Would Establish Incremental Goal-Graph Value?

The proposed architecture contains several mechanisms that can improve outcomes for different reasons. A contract can narrow endpoints; provenance can distinguish data sent to the same endpoint; field scope can preserve unrelated metadata; repair can retain the legitimate part of a mixed request. A goal graph may help locate which restrictions are needed, but it should not receive credit for benefits already supplied by those other mechanisms. This is why B2R is a more

informative comparator than an unmediated model or a whole-Skill blocker.

In this implementation, the runtime verifies every sensitive action regardless of whether the graph predicted it. The graph's incremental role is to enable repair at selected resources. That role may help avoid unnecessary repair opportunities, but it can also lose utility when reconstruction misses a relevant sink. The experiment therefore tests a particular integration of static selection and dynamic filtering. A lack of advantage would not refute all possible graph-based defenses; it would show that adding this selection layer has not justified its cost under these fixtures and comparators.

The repair optimization itself should be interpreted narrowly. Candidate restrictions are predefined, and their costs favor individual sink restrictions over a global block. Neither the candidate set nor the costs are learned from external human preferences. The minimum is consequently a mathematical property of a small enumerated choice set. It is not evidence that the policy is globally minimal, that the program patch changes the fewest semantics, or that every retained action is necessary. Independent task acceptance supplies the relevant empirical check: whether the restricted execution still achieves the requested result.

B. Internal and Construct Validity

The task and Skill analyzers have separated inputs, but both use the same model family, and the fixtures were authored with AI assistance. Shared inductive biases may make interpretation easier than it would be for independently collected tasks. Source-span validation protects against unsupported quotations, not against incorrect interpretations of genuine quotations. The contract compiler can still overgrant or undergrant authority. Component measurements therefore accompany, rather than replace, end-to-end outcomes.

The final outcome oracle is stronger than a defense self-report because it inspects complete state and committed events using independently specified fixture expectations. Nevertheless, its expectations come from the same research design as the fixtures. No second human independently labelled task permissions, paths, or success criteria. The code review used a separate AI agent and found important oracle and accounting defects before formal execution; that is software quality assurance, not inter-annotator validation. We cannot report human agreement statistics or treat AI review as a substitute for them.

The closed action language also changes the construct being measured. Operations are explicit, paths are literal, and provenance is attached by trusted tools. Real scripts can hide effects in subprocesses, dependencies, binary assets, indirect callbacks, or remote services. Here, an unsupported operation becomes an error instead of acquiring an alternate operating-system route. This containment makes the experiment safe and interpretable, but it removes entire bypass classes. A claim of unrestricted agent security would exceed the evidence.

False blocking is sensitive to the request stream. If a method blocks a read, subsequent predicted handles may become invalid, and those malformed requests are not independently legal events. If the model never proposes a necessary action, no denial occurs. FBR can therefore look favorable while task success is poor. Reporting both metrics, together with traces and benign completion, is necessary to distinguish a genuinely selective gateway from one that suppresses useful work indirectly.

C. External and Statistical Validity

The main split holds out named families but shares the base workflow pattern and several mechanisms with development. Two instances vary resource names and contents rather than supplying independent real applications. The three domain labels are contextual descriptions over one substrate. These choices support a controlled mechanism study; they do not establish broad cross-domain generalization. The transfer cohort is also small and changes the original execution environment. Original static bundles provide complementary realism, but classification agreement on text cannot bridge the gap to native execution.

Only GPT-6 Luna is tested. Its behavior depends on the API model returned at execution time, the prompt hierarchy, strict output schema, low reasoning effort, and the two-round budget. Other models, longer trajectories, adaptive attackers, or a less explicit instruction hierarchy may produce different attack rates. Repeating calls captures some sampling variation but does not turn one model into a model population. The returned model identifiers and request records are retained so that future replication can identify version differences rather than silently pooling them.

Family-level resampling respects more dependence than treating every method/repeat as independent. Six clusters nevertheless provide limited statistical resolution, and related templates create dependence even across the named clusters. Reported intervals are descriptive uncertainty summaries. A zero attack count under B0 creates an especially important identification limit: there is no observed susceptibility for an added defense to reduce in that cohort. Stress replay can then demonstrate that an enforcement mechanism responds to imposed actions, but cannot manufacture an empirical live-model improvement.

Finally, original benchmark labels encode the benchmark's intended injection conditions, not an independently resolved permission contract for every selected task. Some contextual actions have legitimate uses. Static false positives and false negatives must be interpreted alongside evidence coverage and policy context. We retain incomplete projections and unresolved analyses rather than treating the acquisition process as an invisible preprocessing step.

D. Practical Implications

A deployable design would need explicit authority bindings, a trusted collector covering every effect, and a way to resolve ambiguous task meaning. The present system's conservative source union prevents easy provenance erasure, but it also

loses precision after aggregation. A useful next implementation would preserve field-level or element-level origins through transformations so that authorized parts can be recovered without guessing. That improvement must be tested against both under-tainting attacks and over-tainting utility failures.

Deployment would also require a different response to unresolved authority. This study fails closed automatically because its fixtures have a finite reference specification and no human intervention. In a real workflow, a carefully scoped request for clarification may be preferable to either silently widening access or abandoning the task. Such an interaction would change latency, task success, and user burden, and deserves its own evaluation. We make no claim that the fully automatic protocol establishes the optimal human-facing policy.

VII. Conclusion and Future Work

This study implemented and evaluated goal-aware capability reconciliation for a bounded Skill-agent setting. The design separates current-task authority from untrusted descriptions of Skill behavior, anchors model-generated proposals to source evidence, and mediates structured effects with deterministic source and field checks. It also distinguishes static potential paths from runtime effects and task completion. These separations make failures inspectable: a denied legitimate action can be traced to an undergranting contract, a missed repair opportunity to graph selection, and an incomplete upload to the granularity of the available data filter.

The experiment produced 1,920 valid executions under a frozen schedule. Its most important result is a qualification of the proposed method's value. In live execution, the unmediated Luna agent did not perform the tested attacks and achieved 99.3% task completion in the main cohort. Additional mediation therefore had no observed attack rate to reduce, and the proposed method completed fewer tasks. In stress replay, provenance and field checks prevented effects that a destination-only gate missed. Selective repair improved attacked joint success from 33.3% with a deny-only gateway to 83.3%. However, the task-only selective repair comparator matched the goal-targeted method. The evidence supports the filtering mechanism in this environment, while leaving the incremental benefit of static goal selection unestablished.

This conclusion matters for experimental design. A complex security pipeline can appear effective when compared only with no defense or a coarse blocker. Adding a comparator with the same enforcement and repair power changes the explanation of that gain. Likewise, reporting only attack success would obscure the cost of rejecting legitimate work, while reporting only task success would miss executions that produce a useful artifact and a forbidden side effect. Joint outcomes and trace-backed acceptance are needed to separate those cases. The present negative comparison is therefore a useful result even though it does not confirm the motivating hypothesis about goal graphs.

Several limits define what remains unresolved. The controlled cases share broad templates across the development and test partitions. Their domain labels do not represent native application integrations, and the transferred attack texts execute under changed tasks. The closed tool language exposes effects and provenance more directly than real programs do. Task interpretation still depends on a language model, and source-span validation does not prove semantic entailment. The reference contracts and acceptance criteria are authored specifications without independent human annotation. These limits preclude claims of general-purpose agent security, native benchmark reproduction, or globally minimal repair.

The first next step is an independently reviewed corpus with genuinely distinct workflow templates and native task acceptance tests. Reviewers should annotate task authority and principal data paths before seeing model predictions, document disagreements, and preserve ambiguous cases rather than forcing binary labels. A subsequent evaluation should retain the strong B2R comparator and include additional models and adaptive attacks. This would test whether graph selection has value when workflows are larger, effects are less explicit, or repair opportunities are more numerous than in the present fixtures.

A second step is to improve the precision of data handling. Whole-handle provenance prevents a simple laundering attack but cannot recover an authorized component after public and private records are concatenated. Element- and field-level provenance, with verified transformation rules, could preserve more utility. That extension needs adversarial tests for incorrect declassification and tests for legitimate transformations; otherwise a higher completion rate could be purchased by silently weakening confidentiality. Repair costs should also represent measured utility loss or explicit user preferences rather than only a fixed count of restricted sinks.

Finally, a native deployment needs a trusted effect collector that covers subprocesses, dependencies, and remote APIs, together with a clear policy for unresolved authority. An interactive clarification may be appropriate when the task is ambiguous, but its burden and latency must be measured. The present artifact supplies a reproducible starting point for those extensions: it records what was proposed, what was allowed or repaired, what actually changed, and which required outcomes remained unmet. Further work should preserve that evidential discipline while expanding the environment, rather than extending the security claim ahead of the measurement.

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